Mark Lodewijk Helwig

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EDUCATION

University of Texas at Austin Graduating May 2027
B.S. Electrical and Computer Engineering Honors **GPA: 3.88**

- Relevant Courses: *Aerial Robotics, Robot Mechanism Design, Automatic Control, Algorithms, Digital Signal Processing Theory, Probability and Random Processes Honors*
- Current Courses: *Human-Centered Robotics, Computer Vision*

WORK EXPERIENCE

Persona AI, *Locomotion Engineering Intern* (Pensacola, FL) May 2026 - August 2026

- Developed, trained, and deployed high-contact behaviors for harness-free robot deployment.
- Developed locomotion policies and deployed them on three different robot hardware platforms.
- Built accurate, compute-efficient robot contact models to reduce the RL sim2real gap.
- Owned the integration of third-party hardware with the company's control framework.
- Debugged hardware and software faults on company robots.

Human Centered Robotics Group, *Researcher* (Austin, TX) September 2024 - Present

- Exploring improvements to high-frequency contact harmonics for Reinforcement Learning.
- Researching discriminator-free style transfer for expressive locomotion in humanoids and quadrupeds using RL in Isaac Lab to deploy on hardware.
- Developed a custom retargeting algorithm for teleoperation of a robotic arm and hand.
- Developed basic whole-body locomanipulation for the Unitree G1 and R1 with sim2sim results.
- Helped develop embedded systems, including CAN hardware and firmware, a ROS2 framework, and an impedance controller, for a novel robotic hand to be controlled via imitation learning.
- Tested and implemented 6-axis F/T sensors in humanoid ankles for multi-contact planning.

Lockheed Martin Aeronautics, *Systems Engineering Intern* (Fort Worth, TX) May 2025 – August 2025

- Developed and owned a Python app for automation of sustainment engineering interpretation of F-35 flight data across all Mission System subsystems that improved process workflow by 92%.
- Owned a codebase with a GUI, automated database interaction (SQL), and data visualization.

Pike Robotics, *Electrical Engineer* (Austin, TX) September 2023 - September 2024

- Worked independently to design and validate an autonomous embedded system that manages an umbilical line and communicates with a host robot in an oil storage tank environment.
- Designed and implemented the main power distribution PCB at 250W, ensuring UL60950-1 general electrical safety standard under UL 1203 C1D1 explosion-proof environment conditions.

PROJECT EXPERIENCE

Quadrotor Autoomy Stack (Austin, TX) Jan 2026 - May 2026

- Built a quadrotor autonomy stack in MATLAB and C++: dynamics simulator, cascaded PD controller, and a UKF fusing IMU, GNSS, and camera data to 5–10 mm accuracy.
- Owned 3D path planning — A* over voxel grids with obstacle inflation, plus a P4 polynomial trajectory generator with bisection timing optimization.

Compliant Finger Mechanism (Austin, TX) Jan 2026 - May 2026

- Designed an under-actuated two-finger gripper as a five-bar linkage with a torsional spring, analyzed as two sequential four-bars so a single servo drives both rigid and wrapping grasps.

SKILLS AND ACCOMPLISHMENTS

- **Software**: Python, C/C++, SQL, Git, Java, MATLAB, Mathematica, ARM Cortex-M4F
- **Software Packages:** Isaac Lab, MuJoCo, PyTorch, ROS 2, OpenCV, TensorFlow
- **Hardware**: KiCad, Solidworks, UART, CANbus, SPI, I2C, EtherCAT, LTSpice
- **Languages**: Native English, Native German, Native Russian, Advanced Spanish
- **Awards/Clearances:** DOD Security Clearance (S), Dr. Beer Endowment for Student Excellence